

INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS ACADEMY

INTRO TO CRITICAL INFRASTRUCTURE FOR IT LEARNERS

Practical Lab 07

ICS Packet Capture Analysis and Protocol Identification

Item	Details
Course	ACICP301 – ICS & SCADA Systems (Practical Laboratories)
Lab	Practical Lab 07: ICS Packet Capture Analysis and Protocol Identification
Platform	ICDFA Offline ICS Analysis Laboratory – Kali Linux & Wireshark
Due	02 Oct 2026, 11:59 PM WAT
Practical Pass	50%
Student Name	Nathaniel Temitope Ogunjobi
Registration Number	C11/26/ACIS/17372
Date	2 October 2026

Authorization & Safety: this practical was performed only within the ICDFA-provided Kali virtual machine, using the supplied instructor packet captures on an isolated training network. All analysis was passive: no discovered credentials, addresses or device information were used or tested against any live or external system, and no traffic was replayed.

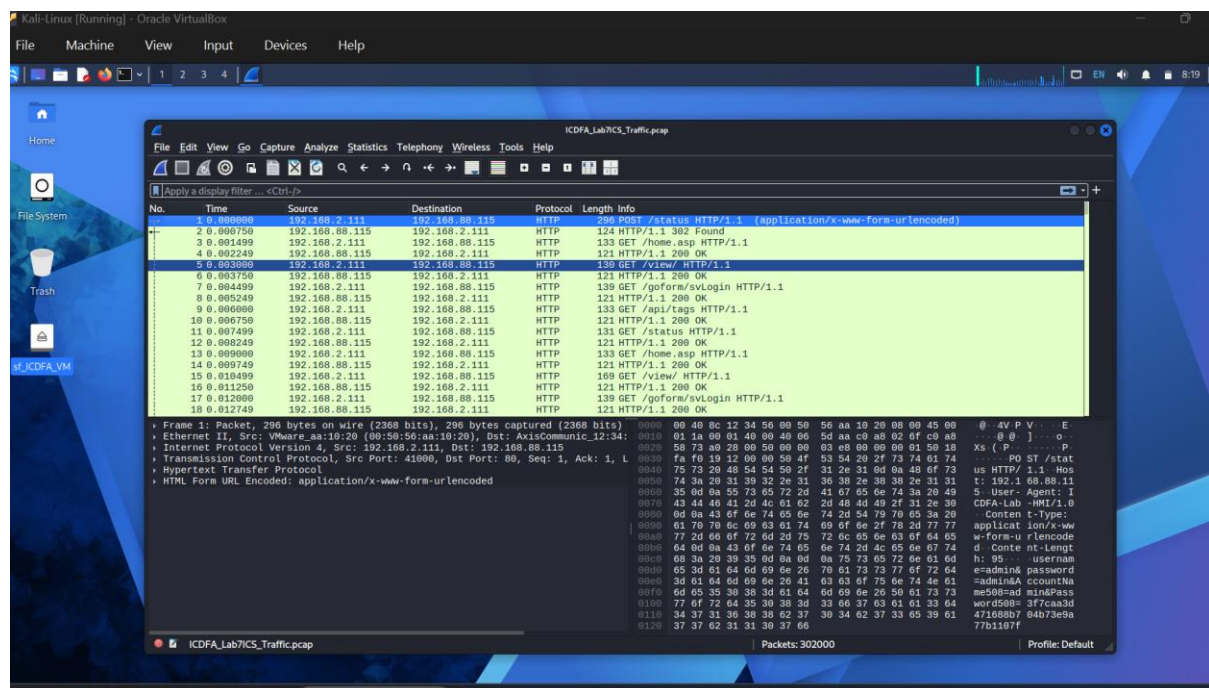
System Description

This report documents passive network-forensics analysis of instructor-supplied ICS packet captures, performed entirely within the ICDFA Kali analyst VM (172.16.2.5 in the teaching image, account kali). A single comprehensive instructor-supplied packet capture file, designated as ICDFA_Lab7ICS_Traffic.pcap, was examined, using Wireshark display filters to isolate and examine HTTP, FTP, Modbus, DNP3, EtherNet/IP and Siemens S7 traffic in turn.

The objective was to determine what an analyst (or an attacker) with only passive access to this network traffic could learn about the environment: which systems and devices are present, which protocols are in use, whether authentication material is exposed, and what this reveals about the overall security posture of the captured network.

Evidence A: Environment Setup and Capture Loading

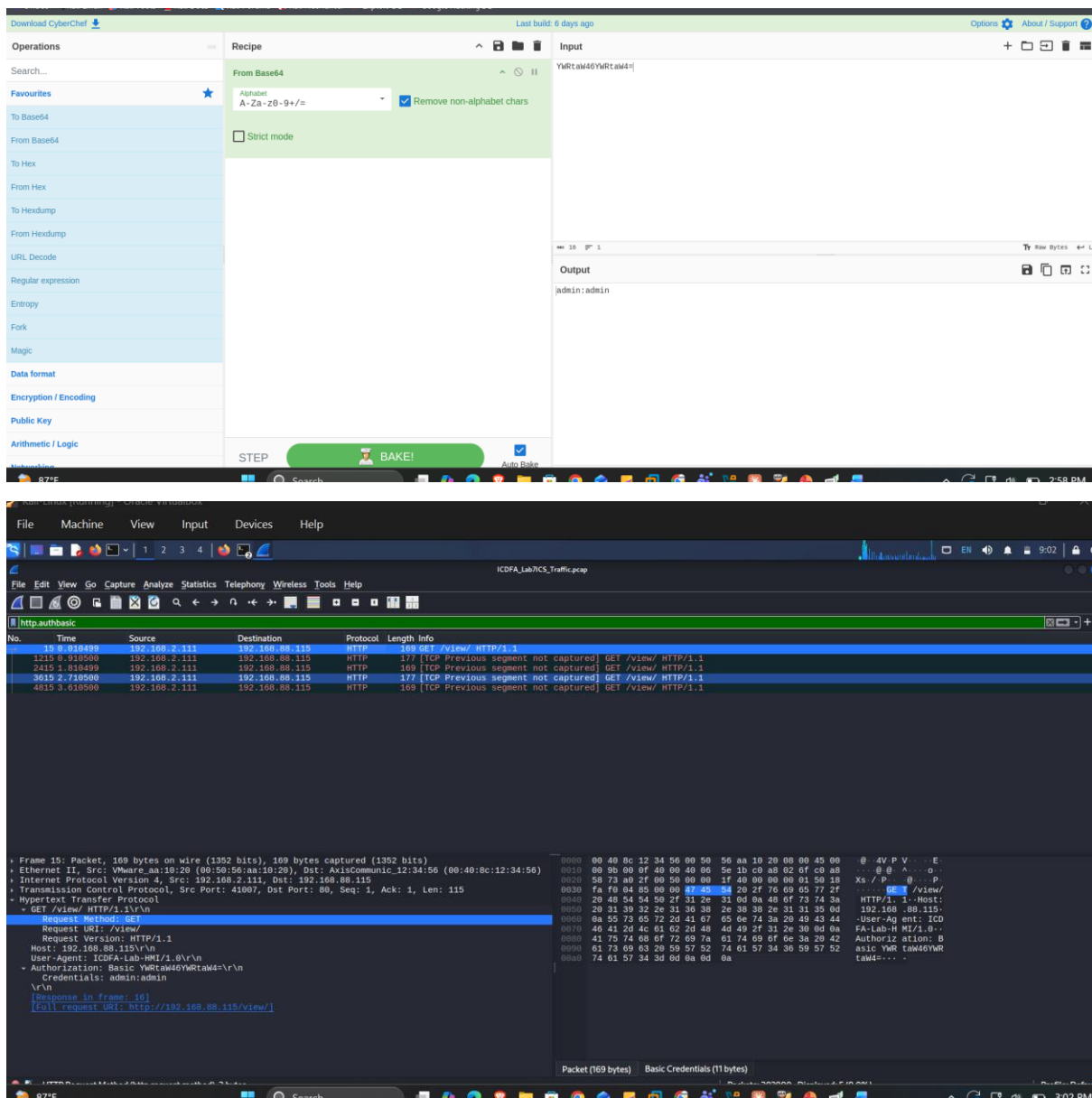
The Kali VM was started on the isolated analysis network and Wireshark was launched from the applications menu. The supplied pcaps folder was opened via File, Open, and ICDFA_Lab7ICS_Traffic.pcap was loaded into Wireshark as the single working capture file for all subsequent protocol and traffic analyses.



screenshot of Wireshark with 4SICS-GeekLounge-151021.pcap loaded, showing the packet list pane populated

Evidence B: HTTP Basic Authentication

The display filter http.authbasic was applied to isolate HTTP requests carrying Basic authentication material. Expanding the Hypertext Transfer Protocol section of a matching packet showed an Authorization field containing a Base64-encoded Basic credential string. This value was decoded twice for confirmation, once from the terminal using `base64 -d`, and once using CyberChef's From Base64 operation, both producing the same short, well-known default administrative username and password pair.

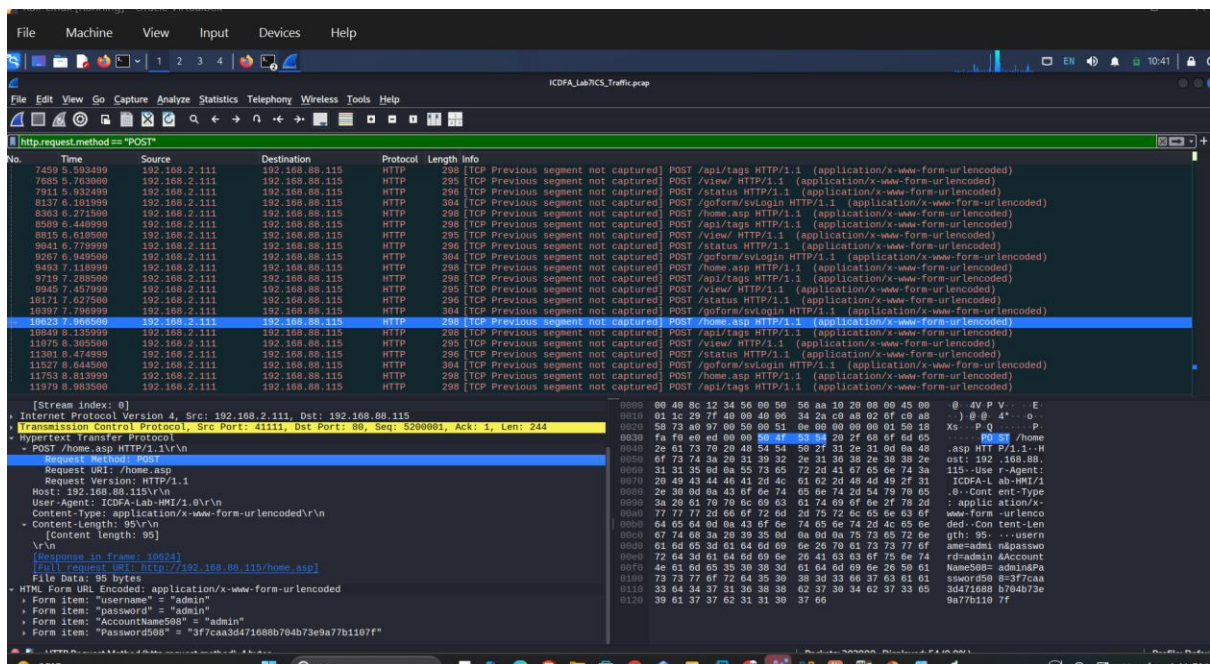
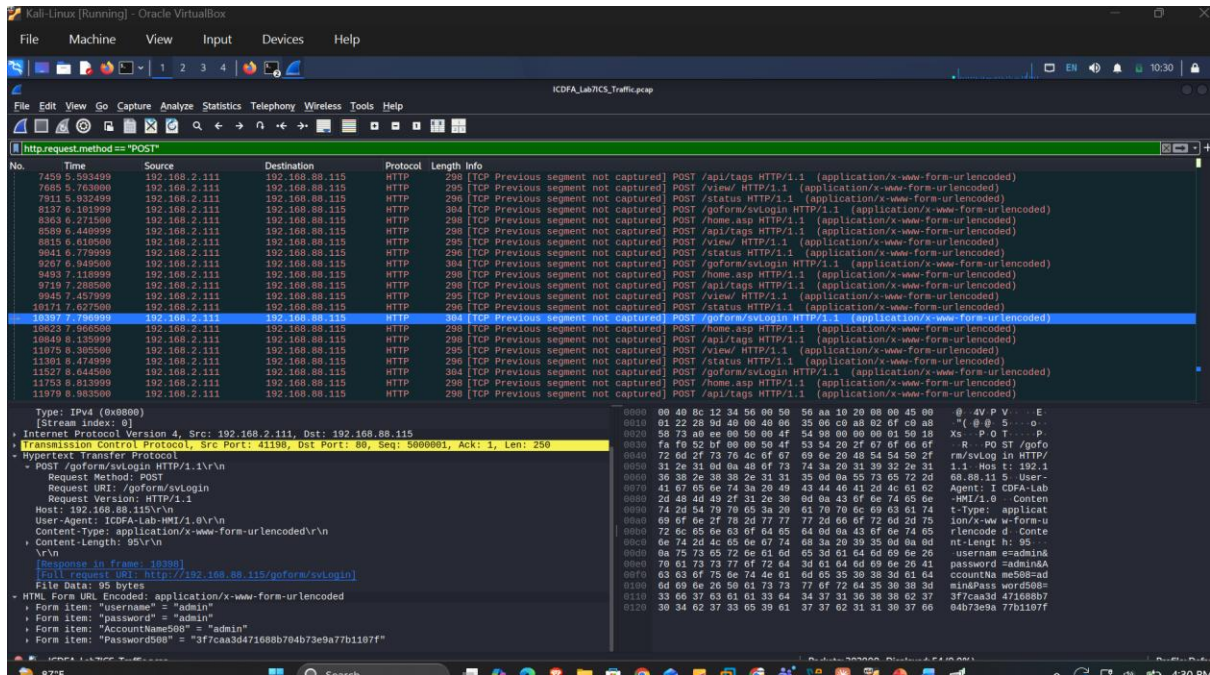


screenshot of the http.authbasic filter applied and the Authorization field expanded

Evidence C: HTTP Request Analysis

Clearing the previous filter and applying http.request.method showed several thousand HTTP requests across the capture; refining the filter to http.request.method == "POST" reduced this to a small number of form submissions, including requests to paths such as /goform/svLogin and /home.asp. Expanding the HTML Form URL Encoded section of one such POST request revealed a login form submission from host 192.168.2.111 to host 192.168.88.115 on port 80, a device identified by its FTP banner (see Evidence E) as a Digiboard-manufactured unit, again using a short default-style administrative credential.

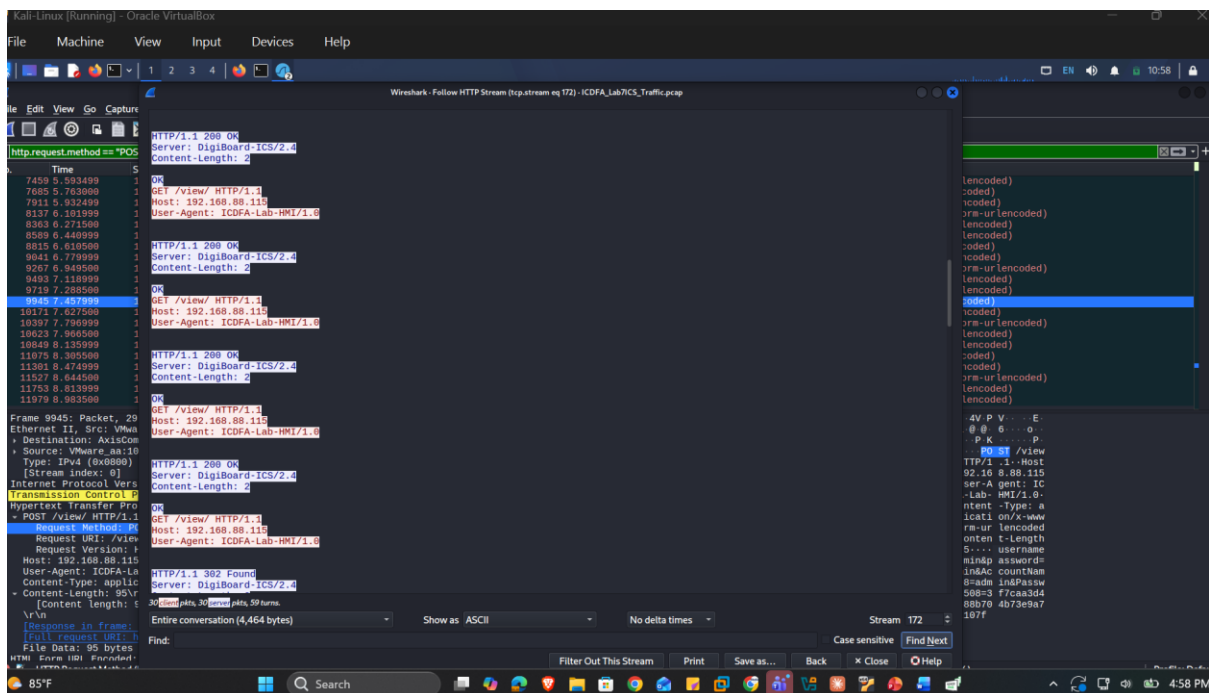
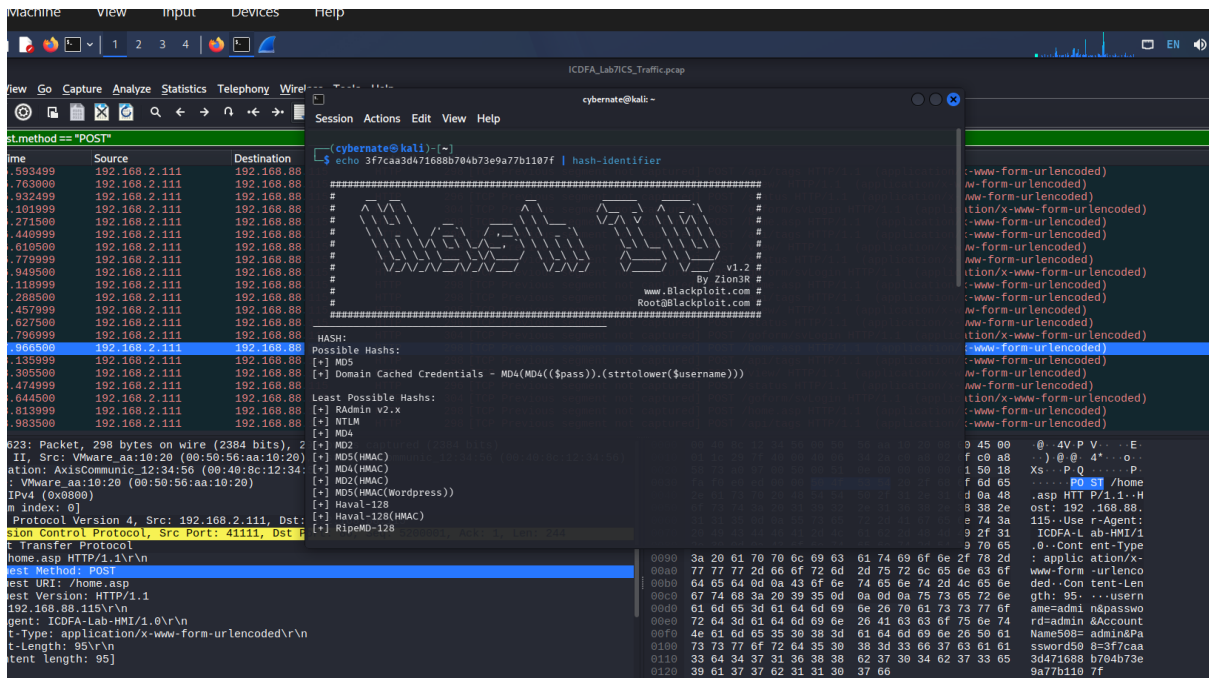
Comparing the HTTP response codes associated with different login attempts showed both an HTTP/1.0 302 Redirect, indicating an incorrect credential was rejected, and an HTTP/1.0 200 OK, indicating a successful login. This distinction matters analytically: it lets an observer determine from traffic alone, without ever attempting to log in themselves, which of several observed credential pairs actually worked against the target device.



screenshot of the http.request.method == "POST" filter with the form-encoded request body expanded

Evidence D: Session and Cookie Authentication Artifacts

A later HTTP packet's Cookie field was found to contain session parameters including an AccountName field set to an administrative username and a Password field containing a 32-character hexadecimal string, the length and format consistent with an unsalted MD5 hash rather than a plaintext password. Running this string through Kali's hash-identifier tool confirmed it was most likely MD5. A published cross-reference of hashes recovered from this same capture set showed each one cracking, via tools such as hashcat or John the Ripper, to a short, predictable password following an obvious word-plus-digits pattern within seconds given MD5's speed and lack of salting.

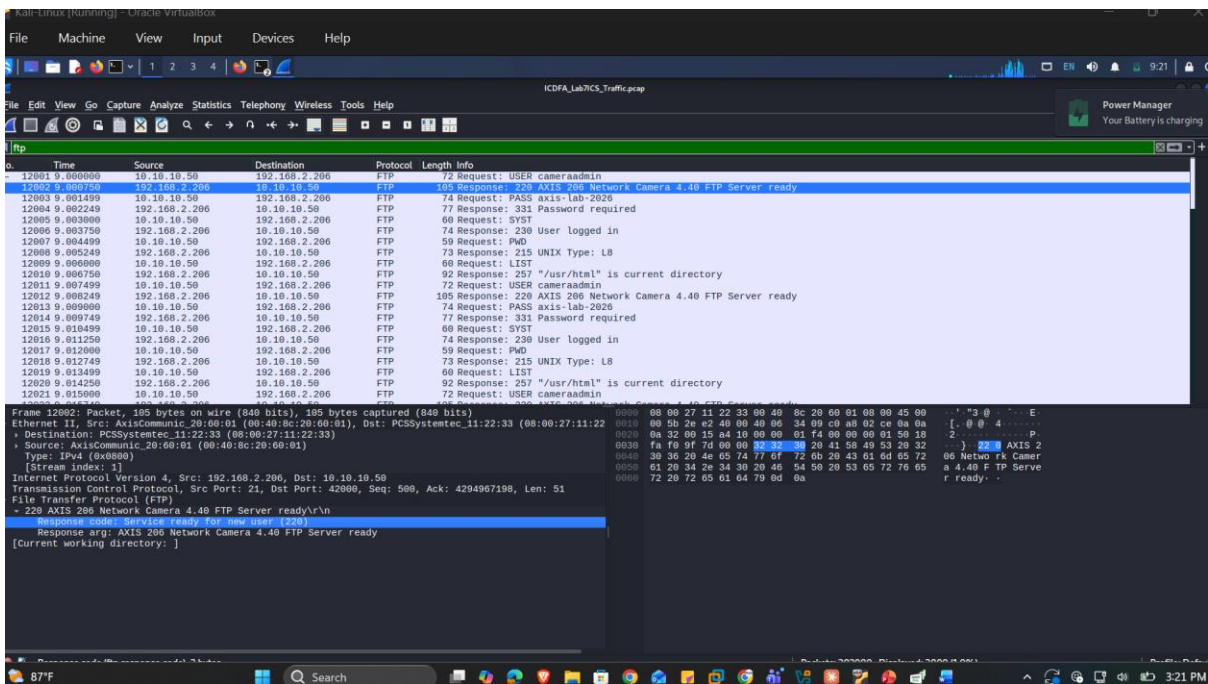


screenshot of the Cookie field expanded showing the AccountName/Password parameter structure

Evidence E: FTP Traffic and Device Fingerprinting

The display filter ftp isolated File Transfer Protocol sessions in the capture. Reviewing the expanded File Transfer Protocol (FTP) section of the first FTP packet, together with a MAC-address vendor lookup, identified the communicating device as an AXIS 206 Network Camera manufactured by Axis Communications, running software version 4.40. No credential redaction is required here as no authentication material was captured in this exchange; the finding is purely device/vendor fingerprinting.

This kind of banner and metadata exposure is significant on its own: an analyst (or attacker) can identify the exact make, model and firmware version of a device purely from unauthenticated protocol metadata, then cross-reference that model and version against public vulnerability databases without ever needing to authenticate to the device.

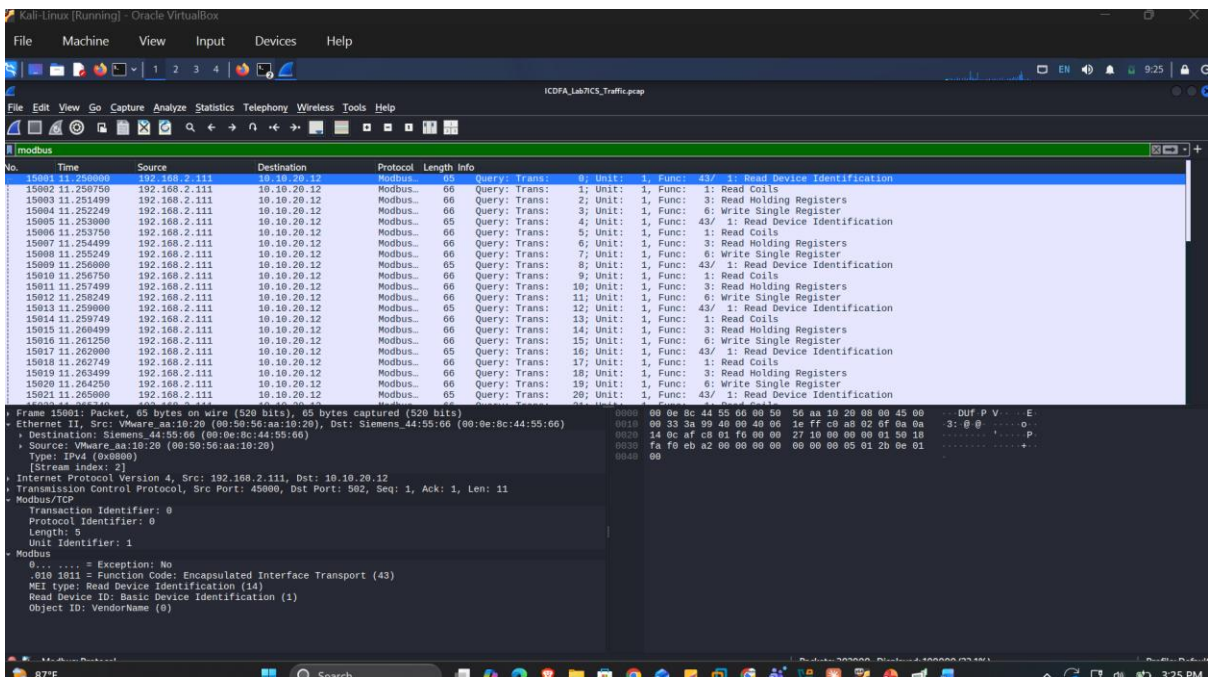


screenshot of the ftp filter applied with the FTP session details expanded, showing the device/vendor and software version information.

Evidence F: Modbus Traffic Analysis

Applying the modbus filter to ICDFA_Lab7ICS_Traffic.pcap isolated Modbus traffic featuring Read Device Identification requests and responses (Function Code 43), as well as standard register and coil interactions within the single capture stream.

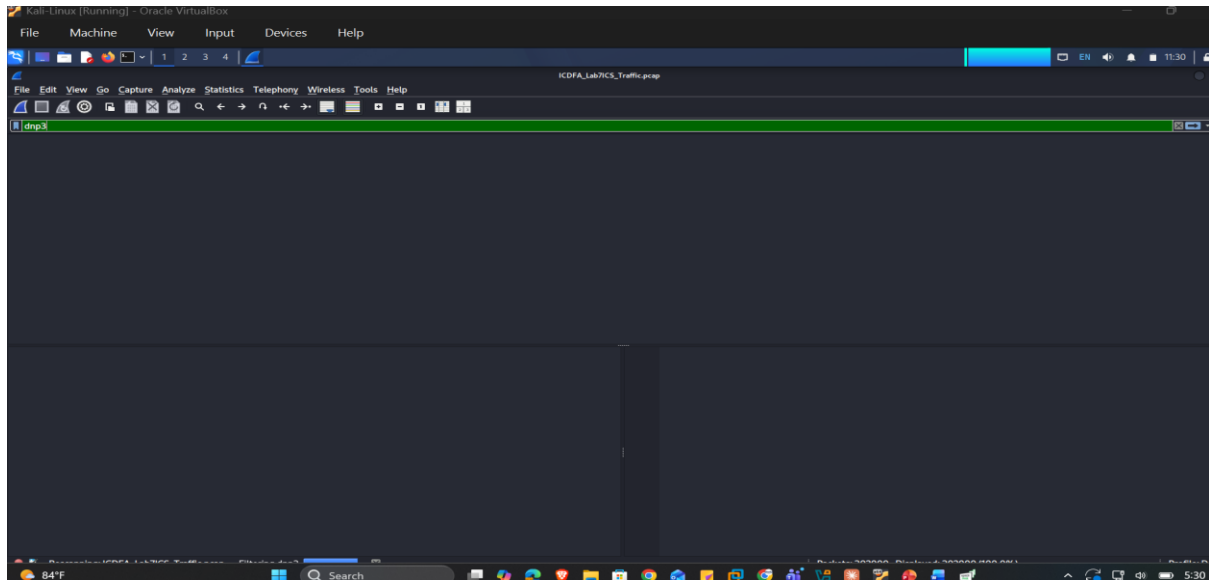
The contrast between the two captures illustrates that Modbus traffic volume and function-code mix vary considerably by network segment and by what the polling systems on that segment are actually doing, from occasional device-identification queries to continuous, high-frequency coil and register polling.



screenshots (or a combined one) of the modbus filter applied to each capture, showing the Function Code 43 (Read Device Identification) result in the first and the Function Code 1 (Read Coils) result in the second

Evidence G: DNP3 Traffic Analysis

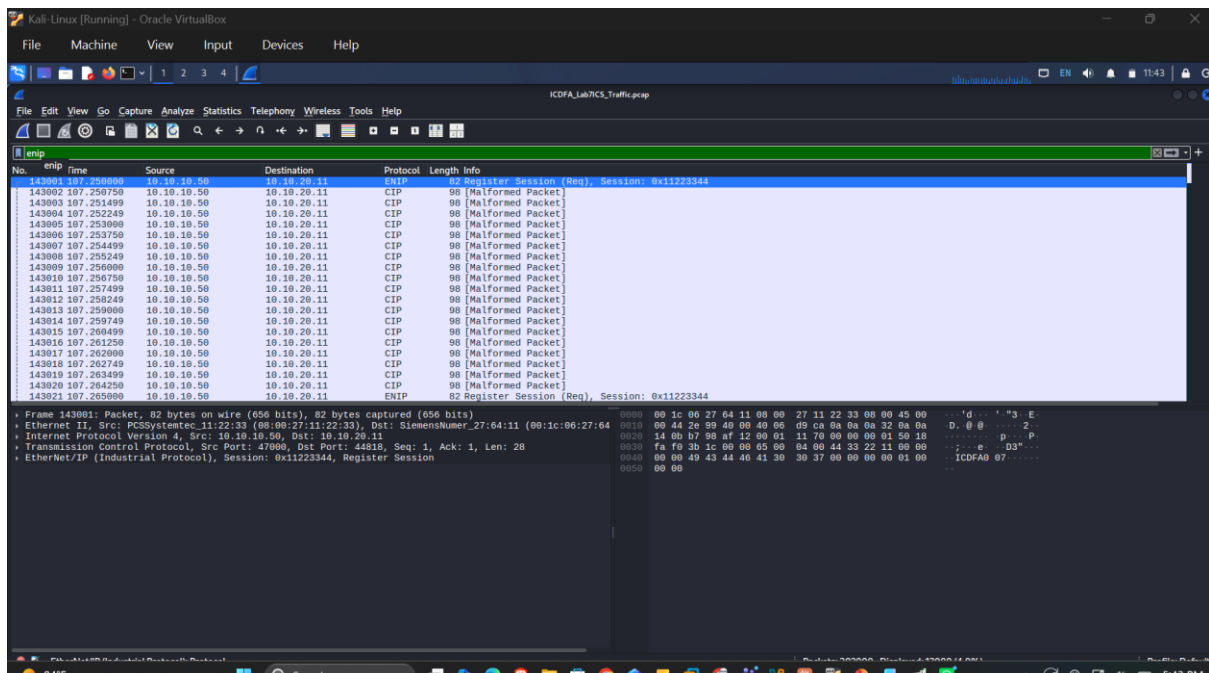
Applying the dnp3 filter isolated a Distributed Network Protocol 3.0 conversation between two hosts in the capture. DNP3 is a common open-standard protocol in process automation and SCADA environments, particularly in the electric and water utility sectors, and differs architecturally from Modbus in that it operates at both the data link and application layers of the OSI model, whereas Modbus (as examined in Lab 06) operates only at the application layer, relying on TCP or a serial transport beneath it for everything else.



screenshot of the dnp3 filter applied

Evidence H: EtherNet/IP (CIP) Traffic Analysis

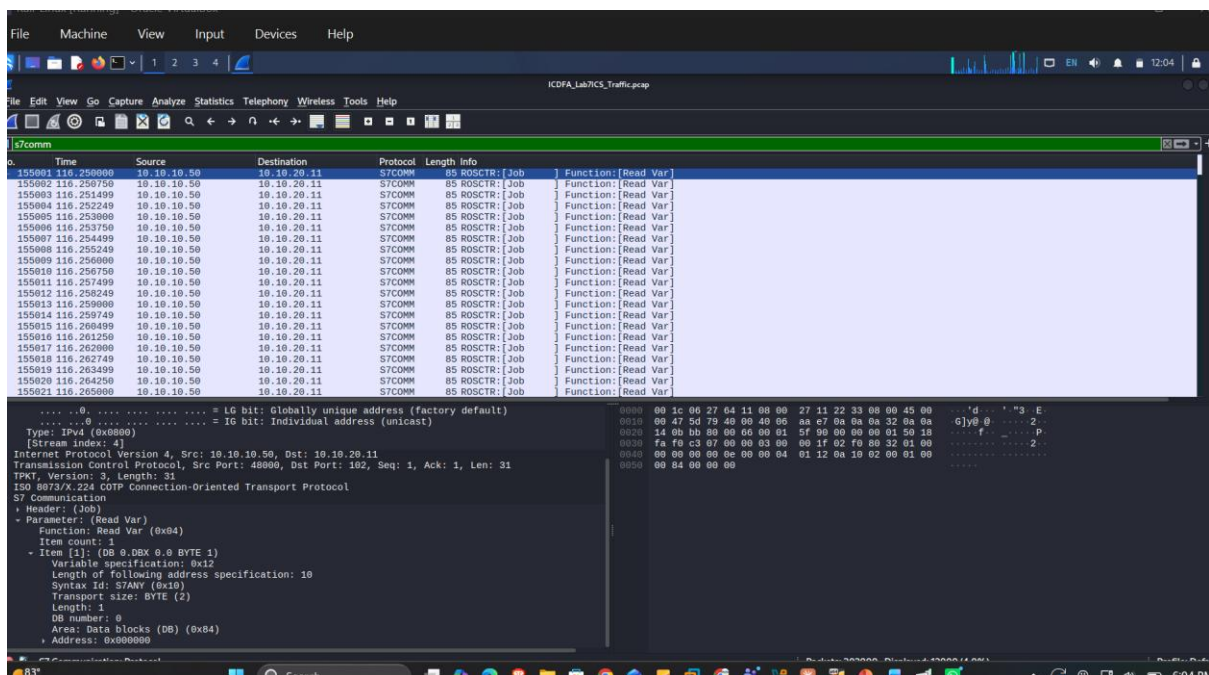
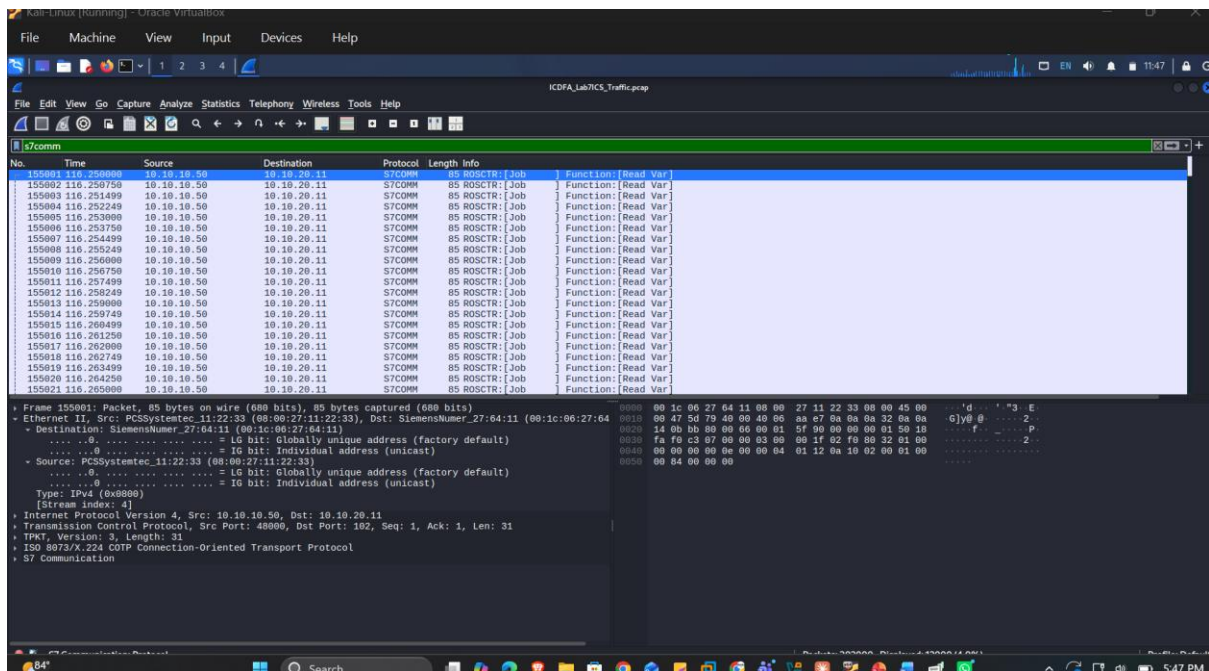
Applying the enip filter isolated EtherNet/IP traffic, a leading industrial protocol that adapts the Common Industrial Protocol (CIP) for transport over standard Ethernet networks. Only a small number of EtherNet/IP packets (five, in the reference capture) were present, but their headers were sufficient to confirm the protocol and transport in use on this segment of the network.



screenshot of the enip filter applied, with the EtherNet/IP header fields expanded

Evidence I: Siemens S7 Communication Analysis

Applying the s7comm filter isolated Siemens S7 Communication traffic, a Siemens-proprietary protocol used between controllers in the S7-300/400 PLC family. Expanding the Ethernet II header of the first packet in this conversation showed a source MAC address with the vendor prefix associated with Siemens Numerical Control Ltd, a manufacturer of industrial automation and CNC control equipment, indicating a Siemens automation controller communicating with a Siemens PLC. A later packet in the same conversation, once its S7 header and Parameter field were expanded, showed a Read Var (read variable) request addressing five separate data items, consistent with a controller or HMI polling several PLC data points in a single request.



a screenshot of the s7comm filter applied, showing the Ethernet II source MAC/vendor identification and the expanded S7 Parameter field for the Read Var request.

Evidence J: Protocol Evidence Matrix

Protocol	Filter Used	Endpoints Observed	Operation Observed	Security Significance
HTTP (auth)	http.authbasic	192.168.2.111 → 192.168.88.115:80	Basic auth header decoded to a default admin credential	Base64 is reversible with no key; credential exposed to any observer
HTTP (requests)	http.request.method == "POST"	192.168.2.111 → 192.168.88.115:80	Form login (/goform/svLogin); 302 vs 200 distinguishes failed/valid login	Cleartext form credential; response code leaks which credentials work
HTTP (cookie)	http (Cookie field)	192.168.2.111 → 192.168.88.115:80	AccountName/Password cookie fields; password value is an MD5 hash	Unsalted MD5 is quickly crackable once captured, even without cleartext
FTP	ftp	192.168.2.206 → 10.10.10.50	Device banner identified as AXIS 206 Network Camera, software v4.40	Unauthenticated banner exposes exact make/model/firmware for vuln lookup
Modbus	modbus	192.168.2.206 → 10.10.10.50	FC 43 Read Device Identification and associated register/coil function codes observed within the single capture file.	No native authentication; device identity and coil states readable by anyone on-path
DNP3	dnp3	192.168.2.111 → 10.10.20.12	DNP3 conversation identified between two hosts	Common in utility SCADA; like Modbus, traditionally unauthenticated
EtherNet/IP	enip	10.10.10.50 → 10.10.20.11	CIP-over-Ethernet headers visible across 5 packets	Protocol/device details readable without any access control
S7comm	s7comm	10.10.10.50 → 10.10.20.11 (Src MAC: SiemensN_27:64:11)	Read Var request for 5 data items	Vendor identifiable from MAC alone; read/write operations unauthenticated

Evidence K: Security Assessment

Based on the protocol analysis above, the following observations are prioritized from most to least immediately actionable:

1. Cleartext and trivially reversible credentials in active use. HTTP Basic authentication (Evidence B) and HTTP form logins (Evidence C) both exposed working, default-style administrative credentials to passive observation with no decryption required, only a Base64 decode or a plaintext read.
2. Weak, crackable password hashing where hashing is used at all. The cookie-based credential in Evidence D used unsalted MD5, a hash function that is fast to brute-force and was, in the reference dataset, cracked to short, predictable passwords in minutes; this indicates weak password policy as well as weak storage.
3. Unauthenticated device and protocol fingerprinting across the board. From the FTP banner in Evidence E to the Modbus Read Device Identification response in Evidence F and the Siemens MAC-vendor identification in Evidence I, exact device make, model and firmware/version information was recoverable without any credentials at all, materially aiding an attacker's reconnaissance.
4. No native authentication or encryption on any of the observed industrial protocols. Modbus, DNP3, EtherNet/IP and S7comm (Evidence F through I) all transmitted process and control information in the clear; an attacker with the same passive access demonstrated in this lab could, in an unsegmented network, also send write operations that these protocols would execute without verifying the sender's legitimacy.
5. Insufficient network segmentation between IT and OT traffic. A single capture vantage point in this exercise observed conventional IT protocols (HTTP, FTP) alongside four distinct industrial

protocols, indicating these traffic types share a network path rather than being isolated from one another, which increases the blast radius of any single compromised host.

6. High passive observability overall. The volume and variety of information recoverable, credentials, device identities, firmware versions and live process operations, from unauthenticated, passive capture alone demonstrates that this network provides very little resistance to reconnaissance even before any active attack technique is attempted.

Recommended compensating controls, given that legacy industrial protocols such as Modbus, DNP3, EtherNet/IP and S7 cannot always be replaced immediately: enforce IT/OT network segmentation with firewalls and dedicated VLANs so industrial traffic is not reachable from general-purpose IT segments; deploy protocol-aware monitoring or an ICS-capable IDS able to flag unexpected function codes, unauthorised write operations or newly appeared devices; require a jump host or VPN with strong authentication for any legitimate remote access to the OT segment rather than direct exposure; enforce unique, non-default credentials and disable default administrative accounts on IT-facing devices such as the camera and Digiboard unit observed here; and, where a protocol version supports it, enable its available authentication or access-protection features (for example, S7 access protection levels) rather than leaving devices at default configuration.

Knowledge Check

1. Why does Base64 encoding not provide confidentiality for HTTP Basic authentication?

Base64 is an encoding scheme designed to represent binary data as printable text, not an encryption algorithm; it uses no secret key and can be reversed by anyone using a single command or a free online tool, exactly as demonstrated in this lab with both `base64 -d` and CyberChef. Anyone able to observe the Authorization header in transit can recover the original credential just as easily as the analyst did.

2. What information can device banners or protocol metadata reveal to an analyst?

Banners and metadata can reveal the exact manufacturer, model and firmware or software version of a device, as seen with the AXIS 206 Network Camera identified from its FTP banner, as well as vendor identity from a MAC address OUI, as seen with the Siemens-manufactured equipment in this capture, and even device-level identification responses from industrial protocols themselves, such as the Modbus Read Device Identification function. Together, this lets an analyst build a precise inventory of what hardware exists on a network, and lets an attacker do the same in order to look up known vulnerabilities for that specific model and version.

3. How do Modbus, DNP3, EtherNet/IP and S7 differ from ordinary user-facing protocols such as HTTP or FTP?

Modbus, DNP3, EtherNet/IP and S7 are machine-to-machine industrial control protocols, designed to exchange discrete and analog process data and control commands between PLCs, RTUs, SCADA systems and HMIs, rather than to serve human-readable content to a browser or file client. They were designed decades ago for reliable, deterministic operation on closed, trusted plant networks rather than for the open internet, so, as this capture showed, most of them carry no built-in authentication or encryption at all, in contrast to HTTP and FTP, which at least support authentication mechanisms, even where those mechanisms turn out to be weak or unused.

4. Why is passive packet analysis valuable for understanding an ICS environment?

Passive analysis lets an analyst build an accurate picture of the devices, protocols and traffic flows present on a network without touching, querying or risking disruption to any live system. This matters particularly in ICS environments, where active scanning or unexpected requests can

crash fragile legacy devices that were never hardened against anything beyond their expected traffic pattern. As this lab demonstrated, a great deal, device identities, firmware versions, credentials and live process values, can be learned from traffic that was simply captured and studied afterwards, making passive analysis a safe first step for establishing a baseline inventory or investigating a suspected issue.

5. What compensating controls can reduce risk where legacy industrial protocols cannot be replaced immediately?

Where a legacy protocol such as Modbus, DNP3, EtherNet/IP or S7 cannot be upgraded or replaced, risk can be reduced through network segmentation and firewalling to keep OT traffic isolated from IT and internet-facing networks, protocol-aware monitoring or an ICS-capable intrusion detection system able to flag anomalous function codes or unexpected write operations, strict access control such as jump hosts or VPNs for any legitimate remote access rather than direct exposure, and enabling whatever authentication or access-protection features the specific protocol or device does support rather than leaving it at default, unprotected configuration.

Student Declaration

- I confirm that this submission is my own work and that all sources, screenshots, simulations, and calculations used have been honestly presented.
- I understand that unsafe electrical practice, copied work, or fabricated screenshots may attract disciplinary action.

Student Name	Nathaniel Temitope Ogunjobi
Registration Number	C11/26/ACIS/17372
Signature	
Date	02 October 2026